Covering Surfaces and Differential Equations

supervised by Dr C. Douglas Haessig

Ryan Lam

The University of Arizona
(Based at the University of Bristol)

Graduate Student Colloquium
February 21, 2023

Introduction

This talk hopes to illustrate the connections between algebraic topology (on complex numbers) and differentiation. In particular, this talks tries to motivate the definition of *Monodromy Representations* on ordinary differential equations (ODEs).
Some textbooks that I like:

- Kuga, Michio. *Galois' Dream: Group Theory and Differential Equations.* Boston, Birkhäuser, 1993.
  - This talk mostly follows Weeks 14-18 of the book.
- Fulton, William. *Algebraic Topology: A First Course.* New York, Springer, 2007.

Table of Contents

1. Algebraic Topology and Where to Find Them
   - Coverings Maps
   - Deck Transformations and Pullbacks
2. Differentiation and ODEs
   - Differentiation and Identifying Functions
   - The Main Result
3. The Upshot (Things I do not understand well)
   - Fuchsian Differential Equations
   - What is My Research About?

Coverings

Let D', D be topological spaces.

Definition

$\pi : D' \to D$ is a **covering (map)** if the following condition holds:

- π is a continuous surjection.
- For any $x \in D$, there is an open neighbourhood U_x such that $\pi^{-1}(U_x) = \bigcup_{i \in \mathcal{I}} V_i$ is a (countable) disjoint union of open sets in D'.
- The restrictions to any open sets $\pi|_{V_i} : V_i \to D$ is a homeomorphism for all $i \in \mathcal{I}$.

Properties of Coverings

Theorem

If D is connected, π is a covering, then for any $x \in D$, the cardinality of $\pi^{-1}(\{x\})$ is the same.

Definition

The elements of $\pi^{-1}(\{x\})$ is called the **conjugates** of x, and its cardinality (independent of x) is called the **degree** of π.

Theorem (Informally)

Given a covering $\pi : D' \to D$ and a curve γ in D, there is a unique lift to a curve $\widetilde{\gamma}$ in D' up to choosing the conjugates of the starting point.

Universal Covering Space

If this screams Galois theory, you are quite right...

Definition

In a covering map $f : \widetilde{D} \to D$, if $\widetilde{D}$ is simply connected, then $\widetilde{D}$ is called the **universal covering space** of D. f is called the **universal covering**.

For "nice" (connected, locally path-connected, semi-locally simply-connected) spaces D,

Theorem

The universal covering space of D always exists and is unique.

Example: Punctured Disk

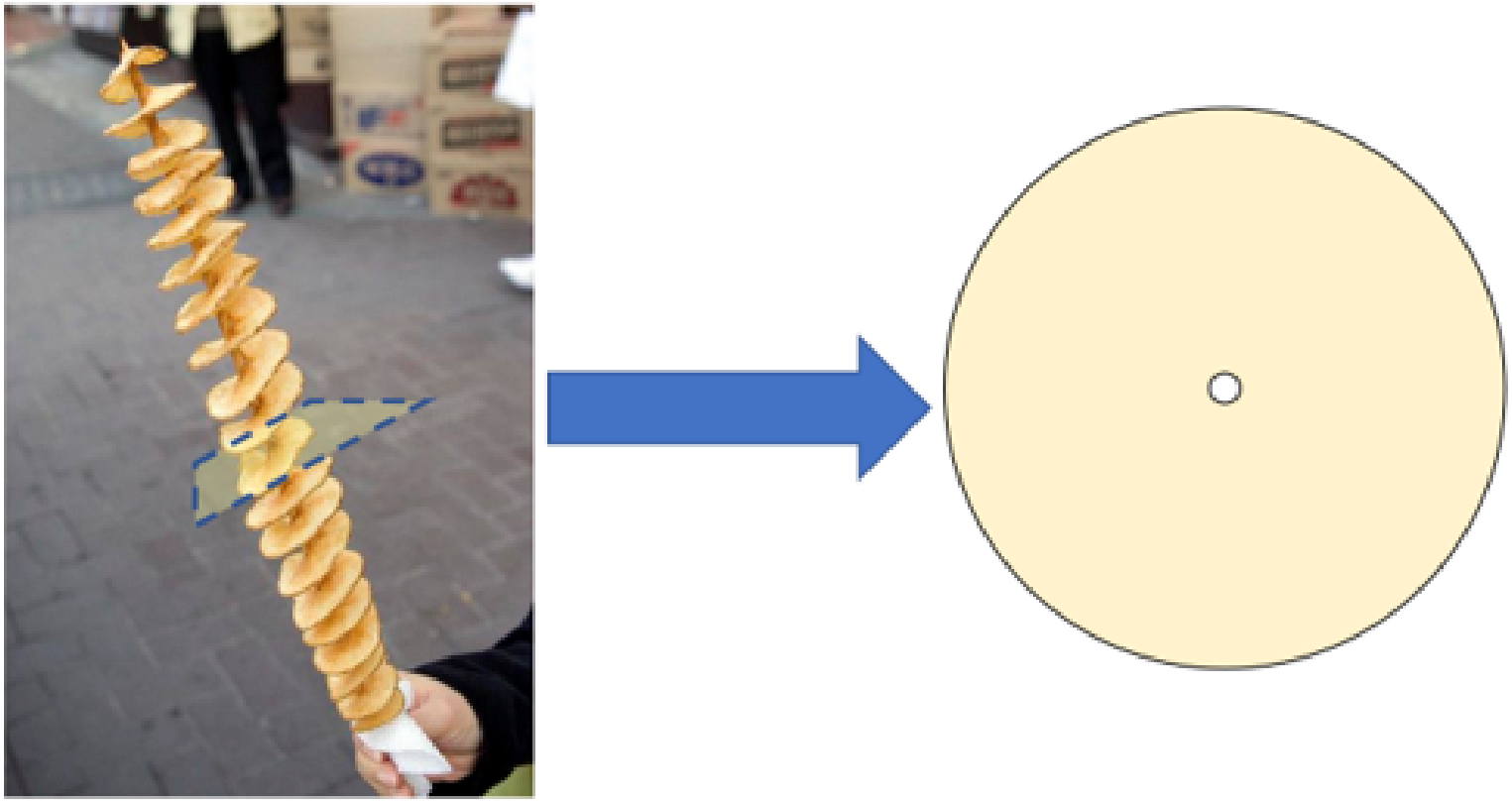


Figure: A "delicious" example demonstrating the universal covering surface of a punctured disk.

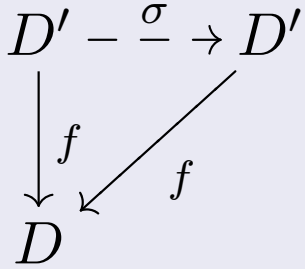
Table of Contents

1. Algebraic Topology and Where to Find Them
   - Coverings Maps
   - Deck Transformations and Pullbacks
2. Differentiation and ODEs
   - Differentiation and Identifying Functions
   - The Main Result
3. The Upshot (Things I do not understand well)
   - Fuchsian Differential Equations
   - What is My Research About?

Covering Transformations

Definition

Given $f : D' \to D$ a covering. A **deck/covering transformation** is a homeomorphism $\sigma : D' \to D'$ such that the following diagram commutes:



If $f : \widetilde{D} \to D$ is the universal covering, the set of all covering transformations is denoted by Γ (or $\mathrm{Aut}(\widetilde{D}/D)$), and it is a group under composition.

Theorem

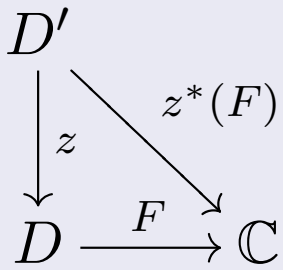
$\Gamma \cong \pi_1(D)$, *the (first) fundamental group of* D.

Pullback of Functions (1)

Now we will call our covering $z : D' \to D$, and we look at a continuous $F : D \to \mathbb{C}$:

Definition

Define $z^*(F) := F \circ z$. So $z^* : C^0(D, \mathbb{C}) \to C^0(D', \mathbb{C})$

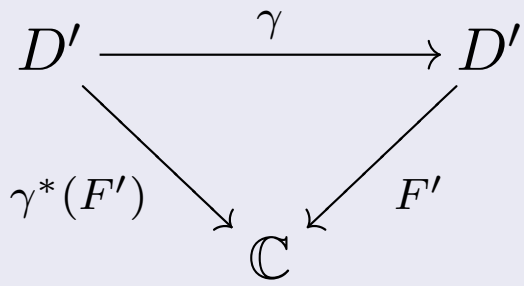


In complex analysis, this is how we distinguish "multi-valued" and "single-valued" functions!

Pullback of Functions (2)

Definition

Similarly for $\gamma \in \Gamma$, define $\gamma^*(F') := F' \circ \gamma$



Both z^* and γ^* are ring homomorphisms.

Example

On the covering $\widetilde{\mathbb{D}} \to \mathbb{D}$, if γ is the "1", then

- $\gamma^*(\operatorname{Log} z) = \operatorname{Log} z + 2\pi i$,
- $\gamma^*(z^\alpha) = e^{2\pi i \alpha} z^\alpha$

Γ-invariant Functions

Definition

A function $F : D' \to \mathbb{C}$ is Γ-**invariant** if $\gamma^*(F) = F$ for all $\gamma \in \Gamma$.

We denote the set of continuous Γ-invariant functions $C^0\left(\widetilde{D}\right)^\Gamma$.

Theorem

If $z : \widetilde{D} \to D$ is the universal covering, then $C^0\left(\widetilde{D}\right)^\Gamma = z^*(C^0(D))$.

Table of Contents

1. Algebraic Topology and Where to Find Them
   - Coverings Maps
   - Deck Transformations and Pullbacks
2. Differentiation and ODEs
   - Differentiation and Identifying Functions
   - The Main Result
3. The Upshot (Things I do not understand well)
   - Fuchsian Differential Equations
   - What is My Research About?

Topological Differentiability

From now on $D \subseteq \mathbb{C}$.
Given $F : D' \to \mathbb{C}$, covering $z : D' \to D$, a point $x_0 \in D'$:

Definition

We say F is **differentiable (with respect to z) at** x_0 if:

- There is a open neighbourhood U containing x_0, a complex number $z_0 \in \mathbb{C}$ such that the function

$$d(y) = \begin{cases} \frac{F(y)-F(x_0)}{z(y)-z(x_0)} & \text{if } y \neq x \\ z_0 & \text{if } y = x \end{cases}$$

  is continuous on U. d is denoted as $\dfrac{dF}{dz}$.

If F is differentiable at all $x_0 \in D'$, we say F is **holomorphic**.

Functions on Covering Surface

Denote all the holomorphic functions on D as $O(D)$.

Theorem

If $f \in O(D)$, $z : \widetilde{D} \to D$ is the universal covering, then:

- $z^*\left(\dfrac{df}{dz}\right) = \dfrac{d(z^*(f))}{dz}$
- $z^*(O(D)) = O(D') \cap z^*(C^0(D)) = O(D') \cap C^0(D')^\Gamma$.

This gives us a way to look at pullbacks entirely on D'!

Theorem

Moreover, for any $\gamma \in \Gamma, F' : D' \to \mathbb{C}$:

$$\gamma^*\left(\frac{dF'}{dz}\right) = \frac{d\left(\gamma^*\left(F'\right)\right)}{dz}.$$

Conclusion: z^*, γ^* respects differentiation.

Table of Contents

1. Algebraic Topology and Where to Find Them
   - Coverings Maps
   - Deck Transformations and Pullbacks
2. Differentiation and ODEs
   - Differentiation and Identifying Functions
   - The Main Result
3. The Upshot (Things I do not understand well)
   - Fuchsian Differential Equations
   - What is My Research About?

The Part Where everyone despise

Theorem

*Let $K \subseteq \mathbb{C}$ be a **simply-connected** domain, then any homogeneous second-order ODE:*

$$\frac{d^2w}{dz^2} + P(z)\frac{dw}{dz} + Q(z) = 0$$

where P, Q are holomorphic, have a unique holomorphic solution $w(z)$ after specifying $w(z_0), w'(z_0)$ for some $z_0 \in K$.

In other words, the solution w forms a 2-dimensional complex vector space, which we will denote by V.

Setting Up Our Main Results

Now we consider the same equation

$$\frac{d^2w}{dz^2} + P(z)\frac{dw}{dz} + Q(z) = 0$$

But it just lives on a domain D, not necessarily simply connected. We can fix it by "lifting". Consider the equation

$$\frac{d^2w}{dz^2} + z^*(P)\frac{dw}{dz} + z^*(Q) = 0,$$

where $z : \widetilde{D} \to D$ is the universal covering. So this ODE lives in $\widetilde{D}$. What do the solutions look like? By the above theorem, the solution also looks like a vector space, which we call $V_{\#}$. We can apply $\gamma \in \Gamma$...

Main Results

Theorem

If $\gamma \in \Gamma, w \in V_{\#}$, then $\gamma^*(w) \in V_{\#}$.
Moreover, the map $w \to \gamma^* w$ is $\mathbb{C}$-linear.

Theorem

The map $\mathcal{M} : \Gamma \to \mathrm{Hom}_{\mathbb{C}}(V_{\#}, V_{\#})$ defined by:

$$\gamma \to \left(w \to \left(\gamma^{-1}\right)^* (w) \right)$$

Is a group representation.

Definition

$\mathcal{M} : \Gamma \to \mathrm{GL}_2 \mathbb{C}$ is called the **monodromy representation**.

Table of Contents

1. Algebraic Topology and Where to Find Them
   - Coverings Maps
   - Deck Transformations and Pullbacks
2. Differentiation and ODEs
   - Differentiation and Identifying Functions
   - The Main Result
3. The Upshot (Things I do not understand well)
   - Fuchsian Differential Equations
   - What is My Research About?

Importance of the Monodromy Representation

Definition

The ODE

$$\frac{d^2w}{dz^2} + \frac{P(z)}{z}\frac{dw}{dz} + \frac{Q(z)}{z^2} = 0$$

where $P(z), Q(z)$is called a **Fuchsian ODE at** $z = 0$.

Theorem (The Frobenius Method)

There is a "power series" solution near $z = 0$, depending on the monodromy representation of the equation.

There is nothing special about 0 here, we can say the same thing for $z = 1$ and $z = \infty$ (make the substitution $z = 1/t$).

Table of Contents

1. Algebraic Topology and Where to Find Them
   - Coverings Maps
   - Deck Transformations and Pullbacks
2. Differentiation and ODEs
   - Differentiation and Identifying Functions
   - The Main Result
3. The Upshot (Things I do not understand well)
   - Fuchsian Differential Equations
   - What is My Research About?

Hypergeometric ODE

Definition

An **hypergeometric ODE** is an ODE of the form

$$z(1-z)\frac{d^2w}{dz^2} + [c-(a+b+1)z]\frac{dw}{dz} - abw = 0$$

where $a, b, c \in \mathbb{C}$ are parameters.

What I am interested in:

- Find a hypergeometric ODE related to an elliptic surface (The *Picard-Fuchs* Equation),
- Find cusp forms related to the ODE, and
- Investigate the monodromy in the space where the ODE lives.

Thank You!

Any questions?